

Project Design Phase

Solution Architecture

Date	05 July 2026
Team ID	PNT2025TMID12345
Project Name	Credit Card Approval Prediction
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Credit Card Approval Prediction – Solution Architecture Diagram:

Credit Card Approval Prediction — Solution Architecture

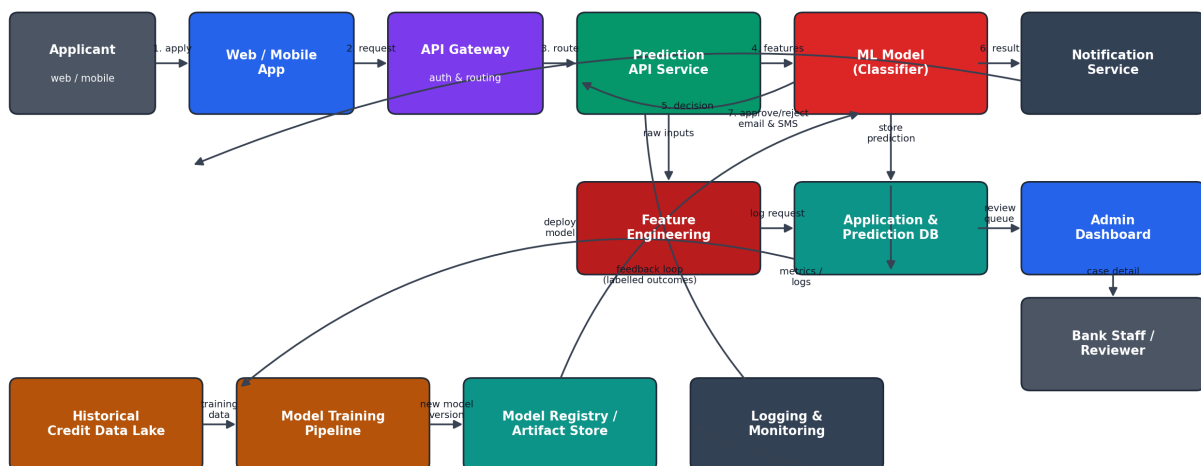


Figure 1: Architecture and data flow of the Credit Card Approval Prediction application, covering the live prediction path (top), request logging and human review (middle), and the model training/retraining pipeline (bottom).

Architecture Walkthrough:

Live prediction flow: An applicant submits details through the web/mobile app. The request passes through an API Gateway (authentication and routing) to the Prediction API Service, which forwards the applicant's data to the deployed ML classifier. The model returns an approve/reject decision, which is sent back through the Prediction API Service and on to the Notification Service, which emails/SMS's the applicant with the outcome.

Logging and human review: Every request and its engineered features are logged in the Application & Prediction Database. Bank staff can review flagged or borderline cases through the Admin Dashboard, which queries the database for case details.

Model training and retraining: A separate offline pipeline pulls historical credit data from the Data Lake, trains and validates candidate models, and publishes approved versions to the Model Registry / Artifact Store. New model versions are deployed to the live Prediction API Service. Labelled outcomes from the production database feed back into the data lake to support periodic retraining.

Monitoring: Logging & Monitoring captures request volume, latency, and prediction drift metrics from both the Prediction API Service and the training pipeline, supporting the performance criteria defined in the functional and performance test plan.